

Learning objective and assessment criteria

Learning objective	Apply one-sample hypothesis testing for proportions and means with correct contextual interpretation at stated significance levels.
Assessment criteria	C4: Applies one-sample hypothesis testing for population proportions in context. C5: Calculates and interprets one-sample mean z -tests with known population standard deviation.

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1 Problem 1: Foundational one-sample proportion test

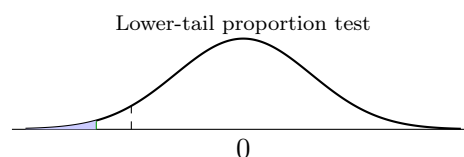
Problem and complete solution (C4)

A district states that 64% of students submit digital homework before the deadline. In a random sample of $n = 300$ students, 174 submitted before the deadline. At $\alpha = 5\%$, test whether the true proportion is lower than 0.64.

- **Parameter:** p = true proportion of students who submit before the deadline.
- **Hypotheses:** $H_0 : p = 0.64$, $H_a : p < 0.64$.
- **Conditions:** $np_0 = 300(0.64) = 192 \geq 10$, $n(1 - p_0) = 108 \geq 10$.
- **Sample proportion:** $\hat{p} = 174/300 = 0.58$.
- **Test statistic:**

$$z = \frac{0.58 - 0.64}{\sqrt{\frac{0.64(0.36)}{300}}} \approx -2.16.$$

- **p-value:** $P(Z \leq -2.16) \approx 0.0154$.
- **Decision:** Reject H_0 since $0.0154 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true on-time digital homework proportion is below 64%.



2 Problem 2: Foundational one-sample mean z -test

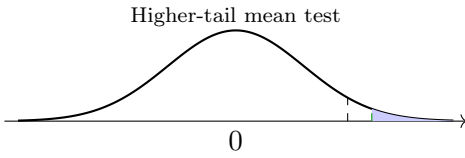
Problem and complete solution (C5)

A transit authority benchmark states that the mean waiting time is 15.0 minutes. A random sample of $n = 100$ riders has mean waiting time $\bar{x} = 15.9$ minutes. Assume known population standard deviation $\sigma = 4.5$ minutes. At $\alpha = 5\%$, test whether the true mean waiting time is higher than 15.0 minutes.

- **Parameter:** μ = true mean waiting time.
- **Hypotheses:** $H_0 : \mu = 15.0$, $H_a : \mu > 15.0$.
- **Test statistic:**

$$z = \frac{15.9 - 15.0}{4.5/\sqrt{100}} = 2.00.$$

- **p-value:** $P(Z \geq 2.00) \approx 0.0228$.
- **Decision:** Reject H_0 since $0.0228 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true mean waiting time exceeds 15.0 minutes.



3 Problem 3: Significance-level sensitivity for a proportion test

Problem and complete solution (C4)

A health office claims that 30% of adults meet a weekly activity target. In a random sample of $n = 250$, 66 meet the target. Test $H_a : p < 0.30$ at both $\alpha = 5\%$ and $\alpha = 1\%$.

- **Parameter:** p = true activity-target proportion.
- **Hypotheses:** $H_0 : p = 0.30$, $H_a : p < 0.30$.
- **Conditions:** $np_0 = 75 \geq 10$, $n(1 - p_0) = 175 \geq 10$.
- $\hat{p} = 66/250 = 0.264$.

$$z = \frac{0.264 - 0.30}{\sqrt{\frac{0.30(0.70)}{250}}} \approx -1.24.$$

- **p-value:** $P(Z \leq -1.24) \approx 0.1075$.
- **At $\alpha = 5\%$:** Fail to reject H_0 , since $0.1075 > 0.05$.
- **At $\alpha = 1\%$:** Fail to reject H_0 , since $0.1075 > 0.01$.
- **Conclusion:** The same sample gives insufficient evidence of a decrease at both significance levels.

4 Problem 4: Directional comparison with fixed evidence (mean test)

Problem and complete solution (C5)

A learning platform benchmark is $\mu_0 = 70$ minutes average weekly study time. A sample of $n = 64$ students has $\bar{x} = 68.0$ minutes, with known $\sigma = 8$ minutes. Use $\alpha = 5\%$.

Tasks

- (a) Test $H_a : \mu < 70$.

- (b) Test $H_a : \mu > 70$.

- (c) Compare the contextual conclusions.

- **Common statistic:**

$$z = \frac{68.0 - 70}{8/\sqrt{64}} = \frac{-2.0}{1.0} = -2.00.$$

- (a) Lower-tail test

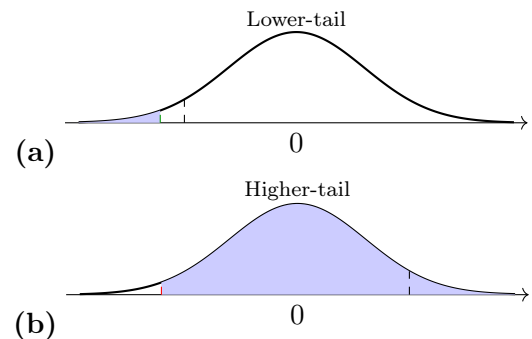
- p-value = $P(Z \leq -2.00) \approx 0.0228$.
- Reject H_0 since $0.0228 < 0.05$.
- There is sufficient evidence that the true mean study time is below 70 minutes.

- (b) Higher-tail test

- p-value = $P(Z \geq -2.00) \approx 0.9772$.
- Fail to reject H_0 since $0.9772 > 0.05$.
- There is not enough evidence that the true mean study time exceeds 70 minutes.

- (c) Comparison

- The data are below the benchmark, so evidence supports the lower-direction claim but not the higher-direction claim.



5 Problem 5: Synthesis across test framing and thresholds

Problem and complete solution (C4)

A customer-service center reports that 70% of calls are resolved on first contact. A random sample of $n = 400$ calls shows 292 resolved on first contact.

Tasks

(a) Test $H_a : p > 0.70$ at $\alpha = 5\%$.

(b) Test $H_a : p < 0.70$ at $\alpha = 5\%$.

(c) Test $H_a : p \neq 0.70$ at $\alpha = 5\%$ and $\alpha = 1\%$.

- **Parameter:** p = true first-contact resolution proportion.

- **Common setup:** $\hat{p} = 292/400 = 0.73$, $p_0 = 0.70$.

- **Conditions:** $np_0 = 280 \geq 10$, $n(1 - p_0) = 120 \geq 10$.

- **Test statistic:**

$$z = \frac{0.73 - 0.70}{\sqrt{\frac{0.70(0.30)}{400}}} \approx 1.31.$$

(a) **Higher-tail at 5%**

- p-value = $P(Z \geq 1.31) \approx 0.0951$.

- Fail to reject H_0 since $0.0951 > 0.05$.

(b) **Lower-tail at 5%**

- p-value = $P(Z \leq 1.31) \approx 0.9049$.

- Fail to reject H_0 since $0.9049 > 0.05$.

(c) **Two-tailed**

- p-value = $2P(Z \geq 1.31) \approx 0.1902$.

- At $\alpha = 5\%$: fail to reject H_0 .

- At $\alpha = 1\%$: fail to reject H_0 .

- **Contextual synthesis:** Across all framings and both thresholds, evidence is insufficient to claim a real change from the 70% benchmark.

